# Victor Angel Lopez Romero

Software Developer | AI Engineering
victorwkey@gmail.com | +52 462 257 4618 | www.victorwkey.com
linkedin.com/in/victorlopezwk | github.com/VictorWKey
Irapuato, Guanajuato, Mexico | Target location: Leon, Guanajuato

## PROFESSIONAL SUMMARY

Software developer specializing in AI engineering, with over one year of professional experience building conversational agents, RAG systems, and LLM workflows for business applications. Focused on understanding business problems and building software that is easy to use, while keeping complexity and resource use in check.

## SKILLS

Generative AI: RAG, AI agents, tool calling, LangChain, embeddings, vector databases, structured outputs, OpenAI, Ollama

ML and NLP: PyTorch, Hugging Face, scikit-learn, pandas, NumPy, BERT fine-tuning, model evaluation, BERTopic

Programming and APIs: Python, TypeScript, JavaScript, Node.js, Express, REST APIs

Data and messaging: SQL, PostgreSQL, Qdrant, Redis, BullMQ

Frontend and tooling: React, Electron, Docker, Git

Familiar with: Context engineering, prompt engineering, multi-agent patterns (agents-as-tools, swarm), agent evaluation, LangGraph, MCP, A2A

## WORK EXPERIENCE

**Vortice Coaching** Oct 2025 - Present
*AI Software Engineer*

- Built a conversational AI agent with tool calling to retrieve information through application APIs.
- Implemented RAG features for document-based question answering, with semantic retrieval and source references.
- Developed LLM workflows for information analysis, summarization, and structured outputs.
- Built backend services and APIs with input validation, access control, and data persistence.
- Implemented process management features and integrations across services; contributed to interface and interaction improvements.
- Reduced token consumption by 42% against the existing baseline, measured in development tests.

## ACADEMIC PROJECTS

**TourlyAI - NLP for tourism reviews | github.com/TourlyAI** Sep 2025 - Mar 2026
*Academic project | Software development and model fine-tuning*

- Built a 9-stage NLP pipeline for tourism reviews, combining sentiment analysis, BERTopic topic modeling, and LLM-generated summaries and insights.
- Fine-tuned and evaluated 2 BERT models with PyTorch and Hugging Face: 12-category multi-label classification (80% weighted F1) and subjectivity detection (77% F1).
- Integrated the pipeline into an Electron, React, and Python desktop app with interactive dashboards and PDF reports, using LangChain for local inference with Ollama and optional OpenAI integration.

## EDUCATION

**Universidad de Guanajuato** Aug 2023 - Nov 2026 (expected completion)
*B.Eng. in Data Engineering & Artificial Intelligence (in progress)*

## CERTIFICATIONS

**AWS Certified Developer - Associate** Oct 2026

**AWS Certified AI Practitioner** Nov 2026

## LANGUAGES

Spanish: Native | English: B1 (Intermediate)